

Journal of Rock Mechanics and Geotechnical Engineering

Prediction of static liquefaction landslides in loess: Integrating triaxial laboratory data into sliding-block model

--Manuscript Draft--

Manuscript Number:	JRMGE-D-25-01573
Article Type:	Full Length Article
Keywords:	Static liquefaction; loess landslide; pore water pressure; sliding-block model; packing structure
Corresponding Author:	Fanyu Zhang Lanzhou University CHINA
First Author:	Fanyu Zhang
Order of Authors:	Fanyu Zhang Jianbing Peng Yixiao Zhang Yapeng Wang Tongwei Zhang
Abstract:	Static liquefaction landslides are among the most catastrophic geohazards, causing serious casualties and damages worldwide. The rapid mobility of this kind of landslide is the most spectacular in the Chinese Loess Plateau (CLP). However, it has been very difficult to accurately predict static liquefaction landslides in loess. Here, we conduct a series of undrained triaxial compression tests for the undisturbed and remolded loess samples in CLP, and compile a fully database of undrained triaxial compression tests of saturated loess combining current and published triaxial tests. Then, we analyze the relationship of normalized stress ratio and pore water pressure ratio in a stress state framework to obtain two key fitted parameters at instability and failure points, and the two ratios and the parameter are integrated into the limit equilibrium equation to build a sliding-block model. The model accurately predicts the factor of safety against initiation and failure of nine static liquefaction loess landslides. The evidences from scanning electron microscope images and grain size distribution characteristics show that packing structure affect shear behaviors and steady state locus in the triaxial tests. The pore water pressure and boundary parameters in the landslides are more sensitive than the parameters extracted from the triaxial laboratory in the sliding-block model. Finally, we develop a hydro-mechanical coupling criterion for predicting instability and failure of the static liquefaction landslides. Our findings show that the new sliding-block model bridges from the triaxial laboratory data to the field site prediction, which can serve as an effective tool for static liquefaction landslides prediction under stress state framework.

Dear Editor

Prediction of static liquefaction landslides in loess: Integrating triaxial laboratory data into sliding-block model

Attached please find a copy of the manuscript submitted for possible publication in ***Journal of Rock Mechanics and Geotechnical Engineering***. This manuscript is original and has not been submitted for review or publication elsewhere. The co-authors also agreed to this submission.

Thank you very much for your time and work.

Sincerely yours,

Fanyu Zhang

Highlights:

- We conduct a series of undrained triaxial compression tests for the undisturbed and remolded loess samples and analyze the affecting factors on shear behaviors.
- We extract the two key fitted parameters from a fully triaxial laboratory database under stress state framework and integrated them into a sliding-block model.
- The new sliding-block model accurately predicts the factor of safety against initiation and failure of nine static liquefaction loess landslides.
- The new model bridges from the triaxial laboratory data to the field site prediction serving as an effective tool for static liquefaction landslides prediction

Submitted to *Journal of Rock Mechanics and Geotechnical Engineering* for possible
publication

**Prediction of static liquefaction landslides in loess: Integrating triaxial laboratory
data into sliding-block model**

Fanyu Zhang ^{1, 2*}, Jianbing Peng ³, Yixiao Zhang ¹, Yapeng Wang ¹, Tongwei Zhang ^{1, 2}

¹ Department of Geological Engineering, School of Civil Engineering and Mechanics, Lanzhou
University, Lanzhou 730000, China

² MOE Key Laboratory of Mechanics on Disaster and Environment in Western China, Lanzhou
University, Lanzhou 730000, China

³ School of Geological Engineering and Geomatics, Chang'an University, Xi'an 710064, China

*Corresponding author: Fanyu Zhang (zhangfy@lzu.edu.cn)

Title: Prediction of static liquefaction landslides in loess: Integrating triaxial laboratory data into sliding-block model

Authors: Fanyu Zhang, Jianbing Peng, Yixiao Zhang, Yapeng Wang, Tongwei Zhang

Abstract: Static liquefaction landslides are among the most catastrophic geohazards, causing severe casualties and damage worldwide. The rapid mobility of this kind of landslide is the most spectacular in the Chinese Loess Plateau (CLP). However, it has been very difficult to predict static liquefaction landslides in loess accurately. Here, we conduct a series of undrained triaxial compression tests on undisturbed and remolded loess samples in CLP, and compile a comprehensive database of undrained triaxial compression tests of saturated loess, combining current and published triaxial tests. Then, we analyze the relationship between the normalized stress ratio and pore water pressure ratio within a stress state framework to obtain two key fitted parameters at instability and failure points. The two ratios and the parameters are then integrated into the limit equilibrium equation to build a sliding-block model. The model accurately predicts the factor of safety against initiation and failure of nine static liquefaction loess landslides. The evidence from scanning electron microscope images and grain size distribution characteristics shows that the packing structure affects shear behaviors and the steady-state locus in triaxial tests. The pore water pressure and boundary parameters in landslides are more sensitive to changes than the parameters extracted from the triaxial laboratory in the sliding-block model. Finally, we develop a hydro-mechanical coupling criterion for predicting instability and failure of the static liquefaction landslides. Our findings demonstrate that the new sliding-block model bridges the gap between triaxial laboratory data and field site predictions, serving as an effective tool for predicting static liquefaction landslides within a stress state framework.

Keywords: Static liquefaction, loess landslide, pore water pressure, sliding-block model, packing structure

1 Introduction

Loess is widely distributed in the Loess Plateau and Other regions of the world (Li *et al.*, 2020). Meanwhile, loess landslides are particularly prevalent in the CLP (Zhang *et al.*, 2014), accounting for approximately one-third of landslides in China (Zhou *et al.*, 2002; Peng *et al.*, 2019). The hazard from loess landslides has caused severe casualties and damage (Zhang *et al.*, 2014; Zhang and Huang, 2018; Zhang *et al.*, 2021a). Static liquefaction loess landslides are among the most hazardous in the CLP. Such landslides occur more frequently due to extreme rainfall events and agricultural irrigation activities in the CLP. As a result of their devastating consequences, there is an urgent need to predict loess landslides due to static liquefaction, which is of crucial importance for hazard mitigation and risk assessment in the CLP.

Liquefied loess landslides have obtained significant attention in literatures (Lutenegger, 1981; Ishihara *et al.*, 1990; Derbyshire *et al.*, 1994; Zhang and Wang, 2007; Peng *et al.*, 2015; Peng *et al.*, 2018; Zhuang *et al.*, 2024), primarily because of their long-runout and high-speed mobility features together with catastrophic results (Zhang *et al.*, 2013; Zhang and Wang, 2018). In addition to the earthquake, extreme rainfall and agricultural irrigation are the most common triggers for the static liquefaction landslides occurring in loess sediments (Zhang *et al.*, 2021b; Zhang *et al.*, 2022). Furthermore, excavation and snowmelt can aggravate rainfall-induced liquefied loess landslides (Zhuang and Peng, 2014; Xian *et al.*, 2022). Thus, the static liquefaction loess landslides have become a focus of considerable research effort in recent years.

There has been considerable research into understanding loess static liquefaction and loess
flowslides (Zhang *et al.*, 2013; Xu *et al.*, 2018; Zhang and Wang, 2018; Liu *et al.*, 2019; Liu *et al.*,
2022; Xu *et al.*, 2022), but much of this effort focused on their mechanisms by examining the
mechanical behavior of liquefied loess (Zhang *et al.*, 2021b). Additionally, there have been many
attempts to model the dynamic processes in numerical simulations, field investigations, and
physical flumes (Zhang *et al.*, 2009; Peng *et al.*, 2015; Wang *et al.*, 2015; Kong *et al.*, 2021; Sun
et al., 2021; Zhang *et al.*, 2021b; Zhang *et al.*, 2022; Zhou *et al.*, 2022). However, these efforts
have mainly concentrated on the mobility features or movement mechanisms of liquefied loess
landslides. Few studies have attempted to predict the initiation of deep liquefied loess landslides
in an area using real-time monitoring data (Xu *et al.*, 2012; Qi *et al.*, 2018; Xu *et al.*, 2020b; Kong
et al., 2021). Meanwhile, several researchers have assessed regional failures of shallow liquefied
loess landslides under extreme rainfall events using stability analysis methods (Zhuang *et al.*,
2017; Zhou *et al.*, 2022; Zhuang *et al.*, 2024). In addition, some mega land development projects
adding new advances and challenges for loess geohazards mitigation in the CLP (Zhang and Wang,
2018; Juang *et al.*, 2019). A catastrophic loess mudflow event occurred in a land development
project in Lanzhou, China, resulting in a loss of more than 1.8 million USD and four fatalities
(Zhang *et al.*, 2019). Based on the above studies, we can conclude that the deep and shallow static
liquefaction loess landslides are closely related to pore water pressure. Recently, several studies
have introduced the framework of critical state soil mechanics to analyze the undrained shear path
of saturated loess (Xu and Coop, 2017; Liu *et al.*, 2019; Feng *et al.*, 2021; Liu *et al.*, 2022). Their
research from triaxial tests reveals a notable difference in strengths (i.e., peak strength and steady-
state strength) between the instability and failure of saturated loess. These findings from the triaxial
laboratory data provide information and evidence to understand the liquefied mechanism of loess

82 flowslides; however, triaxial laboratory data still cannot be used to predict static liquefaction
83 landslides in the CLP.

84 Many studies have been conducted on static liquefaction analysis, as it is crucial to predict
85 flowslides more accurately. As suggested by [Olson and Stark \(2003\)](#), static liquefaction analysis
86 involves three primary tasks, including susceptibility assessment, triggering or initiation
87 prediction, and post-triggering failure stability. Generally, four approaches are used to finish the
88 three tasks that is (1) soil index properties, (2) in-situ penetration tests, (3) stability analysis, and
89 (4) laboratory behavior parameters, as concluded by [Zhang et al. \(2022\)](#). The first two methods
90 are suitable for assessing site liquefaction sensitivity, and the last two methods are well-suited for
91 predicting the initiation and failure of static liquefaction landslides. A common approach is to
92 predict that undrained triaxial test parameters can be integrated into a limit equilibrium stability
93 analysis. Classically, [Poulos et al. \(1985\)](#) used undrained steady-state shear strength to calculate
94 the factor of safety against failure of static liquefaction, whereas [Kramer and Seed \(1988\)](#) used
95 peak shear strength required to initiate liquefaction to calculate the factor of safety. Obviously,
96 this has an essential difference in the initiation and failure of static liquefaction landslides ([Castro,](#)
97 [1969](#); [Kramer and Seed, 1988](#); [Lade, 1992](#); [Leong and Chu, 2002](#); [Yang, 2002](#); [Olson and Stark,](#)
98 [2003](#); [Chu and Wanatowski, 2012](#); [Yang et al., 2022](#)). Thus, Kramer and Seed ([Kramer and Seed,](#)
99 [1988](#)) stressed that static liquefaction landslides should be separately predicted for their initiation
100 and failure at a given slope.

101 Given the above difference in strength, [Sadrekarimi \(2020\)](#) defines a triggering excess
102 pore-water pressure ratio ($r_{u,tr}$) as an index to predict the initiation of static liquefaction landslides.
103 In this procedure, the author collects a large dataset of 1632 laboratory shear tests on cohesionless
104 soils and mine tailings. He distinguishes explicitly the instability and failure of static liquefaction

to define the $r_{u,tr}$. Finally, he verifies the model based on the observed pore water pressure ratio from six flowslides in flume tests and one in-situ landslide (Sadrekarimi, 2020). Some researchers have discussed the proposed rule's limited capability to provide a few seconds of warning of failure and its unpredictability in initiating static liquefaction landslides (Reid *et al.*, 2021). Nevertheless, Reid *et al.* (2021) also strongly agree with the pore water pressures as a key in situ monitoring index, as suggested by Sadrekarimi (2020). They still suggested using the in-situ stress ratio preferentially in the existing analysis framework (Reid *et al.*, 2021). In reply to the foregoing discussion, Sadrekarimi (2021) agreed that $r_{u,tr}$ is close to failure and in-situ stress ratio has the potential to track in-situ pore pressure and overburden stress. Meanwhile, it is crucial to the actual instability line to predict the initiation of static liquefaction landslides (Sadrekarimi, 2021). Despite technological and experimental advancements, uncertainties persist in predicting the initiation of static liquefaction landslides. Reid and her colleagues focus on some of these uncertainties and suggest utilizing in-situ information and experimental data, such as pore water pressure and triaxial shear parameters, to reduce these uncertainties (Reid *et al.*, 2022). However, there remains a significant gap regarding how to integrate triaxial laboratory data into a predictive model to assess and predict static liquefaction landslides accurately

This study aims to integrate triaxial laboratory data into a sliding-block model to predict the initiation and failure of static liquefaction in loess. Firstly, we conduct a series of undrained triaxial compression tests on undisturbed and remolded loess samples from eight different sites in the CLP, and then compile a database of undrained triaxial compression tests on saturated loess, incorporating published laboratory data. Secondly, we analyze the relationship between the normalized stress ratio and pore water pressure ratio to obtain two key fitted parameters at instability and failure points, which are then integrated into the limit equilibrium equation to form

a sliding-block model. The model accurately predicts the factor of safety against initiation and failure of nine static liquefaction loess landslides. Thirdly, we analyze factors affecting shear behaviors in triaxial tests using scanning electron microscope (SEM) images and grain size distribution (GSD) characteristics, and discuss the adaptability of the sliding-block model in predicting static liquefaction landslides under a stress state framework. Finally, we develop a hydro-mechanical coupling criterion index for instability and failure of the static liquefaction landslides. Our findings show that the sliding-block model, when integrated with triaxial laboratory data, can serve as an effective tool for predicting static liquefaction landslides in cooperation with field pore pressure monitoring.

2 Materials and methods

2.1 Testing samples

We took six natural samples and one reconstructed sample at seven different sites in the CLP (Fig. S1). The natural sample is naturally deposited loess. The reconstructed sample is a human-compiled loess, representing a pseudo-natural loess for land development projects in the CLP. The basic index properties of all the samples are listed in Table 1. The reconstructed sample has the greatest dry density due to compaction. The grain size curves and Atterberg limits are plotted in Fig. 1. According to the classification suggested by Gibbs and Holland (1960), the samples primarily represent sandy, silty, and clayey loess, respectively.

2.2 Experiment design and specimen preparation

We performed three groups of experiments in undrained triaxial compress tests on loess samples from different sites. A first group of experiments was aimed at studying the effects of

compaction on the shear behaviors of undisturbed and remolded specimens (T_1 to T_8 in Table 2) from natural (S_1) and pseudo-natural loess (S_2) at the Shichuan sampling site. A second group of experiments was aimed at studying the effects of disturbance on the shear behaviors of undisturbed and remolded specimens (T_9 to T_{16} , as shown in Table 2) at two sampling sites (S_3 , Yuzhon). A third group of experiments was aimed at studying the Effect of soil nature on shear behavior of remolded specimens (T_{17} to T_{27} in Table 2) involving $S_4 \sim S_7$ sampling sites (i.e., Xiji, Yan'an, Xi'an and Tianshui).

The undisturbed specimens were trimmed on the blocked loess samples, and the remolded specimens were prepared by the moist tamping method on the target initial dry density and a desired dry density of 1.5 g/cm^3 . The desired dry density is not only suitable for sample preparation, and also produces very slight differences in void ratio. Triaxial compress tests selected two cylindrical sizes with dimensions of 100 mm height and 50 mm diameter and dimension of 76 mm height and 38 mm diameter. The greater is more convenient for field sampling and uniform compaction, and the smaller is more prone to saturation. All the specimens were saturated with the help of CO_2 and de-aired water subjected to back pressure. The degree of saturation was checked by the B value, which was higher than 0.95. Then, the specimens were consolidated under the desired cell pressures and then compressed under undrained conditions using the strain-controlled method. The axial strain was increased at a constant loading rate of $0.01\%/ \text{min}$. According to the Standard for Soil Test Method (1999), compression at each cell pressure was terminated when the axial strain reached 15%. A detailed summary of all the tests is listed in Table 2.

2.3 Database of undrained triaxial compression tests

We compiled a database of undrained triaxial compression tests combining our current tests and published data (Fig. S1). Generally, the shear behavior of saturated loess is very different with that of sands during laboratory tests (Zhang and Wang, 2018). First, loess has no have the same steady-state shear strength at different normal stress or confining pressure. Second, it is difficult to obtained zero value even though at low normal stress or confining pressure. Thus, we cleared tests for zero shear resistance at different confining pressures or above 100 kPa of confining pressure on the published data. Finally, the database includes 172 saturated loess tests at 16 sampling sites, and its detailed information on the triaxial compression tests is listed in Table 1 and Table S1 in supporting information. The collected samples involve sandy, silty, and clayey loess, and include undisturbed and remolded states with an initial void ratio from 0.46 to 1.10, the range of confining pressure is from 50 kPa to 600 kPa. Thus, all the samples can be behalf of typical types of loess in the CLP.

We concluded four styles of shear behaviors based on the database, including softening, limited softening, creep, and hardening (Fig. 2), which are consistent with general undrained shear behavior of sand suggested by Yoshimine and Ishihara (1998). To failure points of shear test of each saturated loess (Fig. 2), we utilized the general suggestion from previous research (Lade, 1993; Yoshimine and Ishihara, 1998). According to the suggestion from Lade (1993), we used peak location to define the instability points of the softening and limited softening behaviors (Fig. 2a). However, there is a lack of a clear definition for the instability points of creep and hardening behaviors of a soils. Referred to the yielding point definition in metal tensile tests, we suggested that their instability points begin the site of plastic deformation (Fig. 2a). So, no matter what shear

behaviors, we can plot their instability line (IL) and failure line (FL) of saturated loess samples, as shown in Fig. 2c.

Based on compiled database and shear behavior types, we extracted pore water pressure (u) at instability point and failure point to calculate their pore water pressure ratio (r_u) of each sample (Fig. 2b), i.e., a ratio of u and major principal stress after consolidation (σ'_{1c}). Meanwhile, we extracted the mean effective principal stress (p') and deviator stress (q) at the instability point and failure point of the static liquefaction, and calculated their gradient (i.e., state stress ratio, M) and relevant angle (i.e., the angle of shearing resistance, ϕ) under stress space (Fig. 2c). In addition, we defined a ratio of mean effective principal stress (p') and minor principal stress after consolidation (σ'_{3c}). The normalized stress ratio (p'/σ'_{3c}) is similar to that state pressure index suggested by Wang *et al.* (2002) and is different from that suggested by Kramer (1988). As Kramer's stress ratio is a normalized shear stress state. Finally, we analyzed the relationship of p'/σ'_{3c} and r_u at instability and failure of all the samples, respectively.

2.4 Development and verification of sliding-block model

2.4.1 Development of sliding-block model

Here, we integrated all the extracted triaxial laboratory data into limit equilibrium equation to produce a sliding-block model (Fig. 3). We first assumed that the mobilized block is fully saturated sliding along an inclined surface. Meanwhile, there are two kinds of water-loading on the sliding block, i.e., water infiltration wetting and groundwater level rising, corresponding to the shallow and deep-seated landslides, respectively (Fig. 3). Furthermore, we calculated the normal and shear stresses of the sliding block using a graphical procedure in Mohr circle. Because any

point in the Mohr circle is a property on behalf of the stress state on plane space (Holtz *et al.*, 1981), it can easily analyze stress changes in different orientations of the material using the limit equilibrium equation.

So, the stress conditions of a sliding block on an inclined surface are defined as:

$$\begin{aligned}\sigma &= W \cdot \cos\alpha = \gamma_{sat} \cdot z \cdot \cos^2\alpha \\ \tau &= W \cdot \sin\alpha = \gamma_{sat} \cdot z \cdot \sin\alpha \cdot \cos\alpha\end{aligned}\tag{1}$$

Where σ and τ represent the normal and tangential stress, W represents the gravity of the sliding block, α denotes the slope angle, γ_{sat} represents the unit weight of saturated loess, and z represents the height of the unit width sliding block, respectively.

For obtaining σ and τ , we formulate the principal stress in the Mohr circle:

$$\begin{aligned}\sigma_1 &= \sigma + \left(\tan\alpha + \frac{1}{\cos\alpha} \right) \tau = \gamma_{sat} \cdot z \cdot (1 + \sin\alpha) \\ \sigma_3 &= \sigma + \left(\tan\alpha - \frac{1}{\cos\alpha} \right) \tau = \gamma_{sat} \cdot z \cdot (1 - \sin\alpha)\end{aligned}\tag{2}$$

Where σ_1 and σ_3 represent the major and minor principal stress. They can be determined by in-situ conditions or controlled by experimental programs, directly. Hence, Eq. 2 affords a linkage between laboratory shear test and practical slope failure.

According to methods suggested by Poulos *et al.* (Poulos *et al.*, 1985) and Kramer and Seed (Kramer and Seed, 1988), we evaluate the factor of safety against initiation and failure of static liquefaction landslides, respectively. Here, they are calculated by the ratio of the undrained shearing strength (i.e., the resistance against sliding stress, S_u) to the tangential stresses (τ) acting along the sliding surface:

$$F_s = \frac{\text{resistance sliding stress}}{\text{sliding stress}} = \frac{S_u}{\tau}\tag{3}$$

According to the definition from the prior researchers (Sladen *et al.*, 1985; Lade, 1992; Ishihara, 1993; Ishihara, 2008), S_u in the Mohr circle can be calculated by:

$$S_u = \frac{M}{2} \cdot \cos\phi \cdot p'$$

$$M = \frac{6\sin\phi}{3 - \sin\phi}$$
(4)

Where M represents the state stress ratio, here ϕ is the angle of shearing resistance at the instability or failure points.

Next, based on the relationship between p'/σ'_{3c} and r_u , and by integrating Eq. 4 into Eq. 3, a new S_u formula is converted as:

$$S_u = \frac{M \cdot \cos\phi}{2} \cdot \frac{\gamma_{sat} \cdot z \cdot (1 - \sin\alpha)}{\lambda} \cdot \ln(r_u)$$
(5)

Where λ is a fitting parameter stemming from laboratory testing data, which is driven from the relationship between p'/σ'_{3c} and r_u . The details are presented in the subsequent result sections, and r_u is an important index related to in-situ or experimental pore water pressure (see Fig. 9). Finally, a novel sliding-block model is built, and the factor of safety against initiation and failure of static liquefaction can be given by the following equation:

$$F_s = \frac{M \cos\phi}{2\lambda} \cdot \frac{1 - \sin\alpha}{\sin\alpha \cdot \cos\alpha} \cdot \ln(r_u)$$
(6)

2.4.2 Verification of sliding-block model

To verify the predictive ability of the sliding-block model, we collected six liquefied loess landslides in flume test, and three field monitoring liquefied loess landslides. The flume tests all simulated rainfall-induced shallow loess flowslides, they observed deformation and pore water pressure, and some of them have volumetric water content measurements. In the three field

landslide examples, the first is a liquefied vent that occurred in a loess check dam, and the second is only initiation without failure. Note that the third has twice failures and once instability at the same sites. We extracted their sliding thickness, slope angle, and major effective principal stress of the nine liquefied loess landslides, and calculated their factor of safety against initiation and failure of static liquefaction using Eq. 7, respectively. We also compared the predicted r_u , the time-lapsed interval from initiation to failure (T_L), and normalized time (T_N) with those observed at initiation and failure of the practical landslides. The T_N is computed following the method suggested by Scoppettuolo *et al.* (2020). All detailed information was listed in Table S2 in supporting information.

3 Results

3.1 Undrained shear behaviors of saturated loess

3.1.1 Effect of mechanical compaction

Fig. 4 shows the Effect of mechanical compaction on shear behaviors of saturated loess specimens at Shichuan sampling sites (S_1 and S_2), including stress-strain curves, pore pressure-strain curves, and effective stress paths. Figs. 4a-4c show the shear behaviors of naturally and compressively undisturbed loess specimens (T_1 and T_2) at 100 kPa confining pressure.

Specimen T_1 shows a typical shear softening behavior with an increase in pore water pressure until peak strength and then a very slight decrease, while specimen T_2 shows an obvious shear hardening behavior with a continuous increase in pore water pressure. Correspondingly, the effective stress paths are two different styles. Specimen T_1 is inclined to the left with increasing shear stress, and finally reaches its peak shear strength (i.e., the instability point). Thereafter, the

path descends towards its steady-state strength (i.e., the failure point), approaching the zero point. Specimen T₂ is first left-upward until the instability point due to pore water pressure generation. After that, the path ascends right-upward, accompanying further shearing without reaching a truly steady-state strength. This is due to the limited shear displacement in triaxial compression tests. Their completely opposite shear behaviors can mainly be attributed to the significant difference in initial dry density between specimen T₁ (1.39 g/cm³) and specimen T₂ (1.79 g/cm³). Meanwhile, specimen T₂ has undergone a structural disturbance due to compaction during filling.

Figs. 4d-4f and Figs. 4g-4i show the shear behaviors of compressively undisturbed (T₃~T₅) and remolded (T₆~T₈) loess specimens at three confining pressures with the same dry density. All six specimens exhibit shear hardening behavior; their shear behaviors are the same as that of specimen T₂. However, there are some differences among them. The deviator stress of undisturbed loess specimens is lower than that of remolded loess specimens at three confining pressures. The pore water pressure of undisturbed loess specimens is higher than that of remolded loess specimens. The effective stress paths initially move leftward and then ascend rightward with increasing deviatoric stress in the undisturbed loess specimens. While the effective stress paths of the remolded loess specimens initially move rightward, then leftward, and finally ascend rightward and upward during shear.

3.1.2 Effect of specimen disturbance

Fig. 5 shows the Effect of disturbance on shear behaviors of saturated loess specimens at Yuzhong sampling site, including stress-strain curves, pore pressure-strain curves, and effective stress paths. Three confining pressures and two initial dry densities were chosen for the testing group (Table 2). Figs. 5a-5c show the shear behaviors of undisturbed loess specimens (T₉~T₁₁),

and Figs. 5d-5f show the shear behaviors of remolded loess specimens ($T_{12}\sim T_{14}$). It can be seen that specimen disturbance has a significant impact on the shear behaviors of saturated loess at the same sampling site. There are three types of undrained shear behaviors, i.e., softening ($T_9\sim T_{10}$), creeping ($T_{11}\sim T_{13}$), and hardening (T_{14}).

For the undisturbed specimens ($T_9\sim T_{10}$) at confining pressures of 100 kPa and 200 kPa, their shear behaviors exhibit a very slight shear softening behavior, with an increase in shear strength until instability, followed by a slight decrease. While their pore water pressures are continuously increasing with shearing under axial loading, their effective stress paths tend first to the left and finally reach the failure line with increasing deviator stress. Compared to the ring shear test results with a constant pore water pressure (Wang and Sassa, 2002; Zhang and Wang, 2018), the specimens cannot obtain a true steady-state. This is derived from the limited displacement constraint in triaxial shear tests. However, it is a common phenomenon in the triaxial apparatus.

The undisturbed specimen (T_{11}) at a confining pressure of 300 kPa exhibits a similar shear behavior to that of the remolded specimens (T_{11} and T_{12}) at confining pressures of 100 kPa and 200 kPa. It indicates that remolded specimens have higher shear strength at relatively low confining pressure. It is attributed to the fact that disturbances increase the packing structure between particles in the remolded specimen. All three have the same trend-effective stress paths, that is, their paths first trend rightward, then leftward, and finally remain almost level. Nevertheless, some differences in shear behaviors still exist. For the undisturbed specimens at 300 kPa of confining pressure, a stronger creeping trend is observed, accompanied by more lasting changes in shear strength and pore water pressure, compared to the two remolded specimens.

3.1.2 Effect of soil nature

Figs. 6 and 7 show the Effect of soil nature on shear behaviors of remolded loess specimens at four sampling sites ($S_4 \sim S_7$), including stress-strain curves, pore pressure-strain curves, and effective stress paths. Three confining pressures and one desired dry density were chosen for the testing group (Table 2). Fig. 6 shows the shear behaviors of silty loess specimens ($T_{15} \sim T_{20}$) at Xiji (S_4) and Yan'an (S_5) sampling sites. Fig. 7 shows the shear behaviors of clayey loess specimens ($T_{21} \sim T_{27}$) at Xi'an (S_6) and Tianshui (S_7) sampling sites. The remolded specimens exhibit two distinctly different types of shear behavior. This means that the Effect of soil nature is also significant in remolding loess.

In silty loess tests, the shear behaviors of remolded specimens show softening at the Xiji sampling site and hardening at the Yan'an sampling site, respectively. In clayey loess tests, the shear behavior of remolded specimens is hardening at two sampling sites. For hardening remolded specimens, Yan'an has the highest shear strength, Xi'an is in the middle, and Tianshui has the lowest. On the contrary, the pore water pressures of remolded specimens are higher at Tianshui compared to Yan'an and Xi'an. As shown in Fig. 1 and Table. The loess at the Tianshui site has finer grains and higher plasticity. This means that a small amount of plastically fine grain increases their strength and restrains their pore water pressure. Indeed, similar results have been observed in ring shear and triaxial shear tests (Wang *et al.*, 2007; Carraro *et al.*, 2009; Zhang *et al.*, 2021b)

It is worth noting in the three groups of experiments that even in undisturbed specimens, not all effective stress paths drop towards the zero point, and they remain far away from it with increasing confining pressure, along with specimen disturbance. This is an entirely different specimen (T1) from the undisturbed one at the Shichuan site. This can be attributed to two factors.

First, there is looser packing at a lower initial dry density. Second, high confining and specimen disturbance cause denser packing in specimens.

3.2 Analysis of hydro-mechanical parameters

We examine the state line in mean effective principal stress (p') and deviator stress (q) space at instability and failure of all the saturated loess samples, as shown in Fig. 8. On the p' - q space, the state stress ratio (M) and the angle of shearing resistance (ϕ) at instability point and failure point can be described with a linear relationship with high credibility. There is a certain deviation for the entire dataset, but the overall trend remains constant with minimal variability. The observation implies that the constant friction angles at the instability state and failure state are completely independent of the shear behavior of saturated loess, regardless of its initial state and intrinsic nature. As a result, loess shows a unique and similar instability line and failure line on the p' - q space. In addition, M_i and ϕ_i are lower on the instability line than M_f and ϕ_f on the failure line. The failure line is equivalent to the critical state line. It is well known that the failure line in p' - q space has constant M_f and ϕ_f in many studies for various saturated soils (Verdugo and Ishihara, 1996; Sadrekarimi and Olson, 2011; Yang *et al.*, 2022; Ahenkorah *et al.*, 2024). Of course, the failure line is not a permanent constant. It can have a greater M_f , typically when soil samples undergo grain crushing due to sufficiently high confinement (Ghafghazi *et al.*, 2014; Shi *et al.*, 2023).

We analyze the relationship between pore water pressure ratio (r_u) and normalized stress ratio (p'/σ'_{3c}) at instability and failure of all the saturated loess samples, as shown in Fig. 9. As we can see, there is a soundly exponential function relationship whether at the instability points or failure points for all saturated loess. Meanwhile, the fitted line at failure has higher credibility ($R^2 = 0.98$)

than that of Instability ($R^2 = 0.92$). The failure or critical state is a more stable state. Although our defined instability point can track the pore water pressure and stress of the triaxial shear tests on saturated loess, its accuracy should be slightly lower than that of the failure point. Even so, we obtain two key fitted parameters (λ). The instability of λ_i is -1.09, and the failure of λ_f is -0.75, respectively.

Moreover, the hardening shear behavior pattern has low r_u at high p'/σ'_{3c} states (Fig. 9). Our observation suggests that the r_u between 0.2 and 0.6 can be sufficient to trigger or initiate loess static liquefaction, which eventually can result in very high r_u values for static liquefaction failure. And, there is a critical value r_u of 0.6 for liquefaction occurrence at failure (Fig 9b), which is consistent with previous undrained ring shear tests of saturated loess (Zhang *et al.*, 2022). The results further confirm that the critical r_u of 0.6 is reasonable for the liquefaction susceptibility assessment suggested by Seed (Seed, 1987).

The p'/σ'_{3c} is perfected to capture the pore water pressure and consolidation stress state of undrained triaxial shear tests for all saturated loess. Consequently, the current method provides a convenient hydro-mechanical coupling pathway for tracking in situ overburden stress and pore water pressure within a stress state framework. As the M , ϕ , and λ can easily be obtained, they can then be integrated into the sliding-block model for calculating the factor of safety of static liquefaction landslides.

3.3 Prediction of static liquefaction landslides

Fig. 10 shows the factor of safety against static liquefaction for loess landslides from six flume tests and three field monitoring data sets. We successfully captured the factor of safety

against the initiation of nine static liquefaction loess landslides and the factor of safety against the failure of eight loess landslides. The factor of safety of the rest landslide is 1.77 at failure; the result is accurate. The landslide did not occur in final failure, although there has been enough artificial rainfall recharge (Chen *et al.*, 2017). There are several interesting notes for the other two in situ landslides. First, Xu *et al.* (2020a) reported data from monitoring a loess check dam, which recorded three initiations and one failure. The first two initiations occurred in shallower depths than the third one. Our model is based on the sliding surface of the third initiation event, i.e., the failure surface. Therefore, our model cannot calculate the first two initiations, which are reasonably well matched with in situ situations. The only deep-seated event is a static liquefaction loess landslide that occurred in irrigation areas due to groundwater rise (Qi, 2017). He monitored one initiation and two failures due to ground surface deformation resulting from changes in the groundwater level. We use the in-situ sliding surface as a calculated boundary. Our model accurately distinguishes the entire deformation process, and interestingly, finds that the first failure is also the initiation of the second failure. This is consistent with monitoring data and field investigation. On the other side, the stress ratio (K_c) is a function with slope (Ishihara, 2008), i.e., $K_c = \sigma'_{3c} / \sigma'_{1c} = (1 - \sin \alpha) / (1 + \sin \alpha)$, which is directly related parameters with the geometrical features of all the loess landslides, relating with its stress states at a given sliding thickness (Fig. 10). Thus, this can easily be used in the practical landslide stability analysis.

We compare the predicted and observed r_u and T_N at initiation and failure for nine static liquefaction loess landslides, as shown in Figs. 11a-d and Figures. S1-S9. It can be seen that all the r_u and T_N predicted by our sliding-block model are well matched with the observed values from the liquefied vents in flume tests and field landslides. No failure solid dot is reasonable, as highlighted by the red dotted line in Fig. 11b. Here, the predicted r_u is a higher value compared

with the observed r_u , because the predicted value is based on its failure state. The landslide does not occur in final failure after initiation (Chen *et al.*, 2017). This implies that the laboratory data-driven sliding-block model can effectively capture the changes in key state parameters, r_u and TN , at different stages of landslide development.

Fig. 12 shows the time-lapsed interval from initiation to failure (T_L). It can be seen that the time interval of all liquefaction loess landslides from six flume tests and two in-situ cases. For the six flume tests, all simulated loess landslides exhibit static liquefaction under artificial rainfall. The minimum T_L value is 8 minutes, while the other five events exceed 1 hour. The relevant shallow case in the check dam takes 56 days from initiation to failure, and the deeper case in the Heifangtai irrigation area takes 176 days. This suggests that, using our sliding-block model, there is sufficient time for early warning, at least in the case of static liquefaction loess landslides. Of course, the long time interval may interfere with early warning of the final failure stage, but pore water pressure monitoring can fill this deficiency, coordinated with our model calculation. As suggested by Reid *et al.* (2021), a predicting method would preferably provide a few seconds of warning before the failure of static liquefaction landslides. This means that the prediction of our model is successful for all the static liquefaction loess landslides. The warning time is relatively short in Sadrekarimi (2020)'s model for predicting those verified landslides because the $r_{u,tr}$ is too close to their failures (Sadrekarimi, 2021). We argue that the initiation prediction of the Sadrekarimi (2020)'s model becomes inevitably a failure prediction. Even so, he still provides a route to drive a predictive method based on laboratory triaxial tests.

4 Discussion

4.1 Factors affecting shear behaviors of saturated loess

The aforementioned laboratory studies have shown that four types of shear behaviors have been observed in the loess shear test dataset. However, the extracted laboratory parameters are almost consistent in the four shear behaviors. Therefore, it is necessary to provide an analysis of why the extracted laboratory parameters can be integrated into models for correctly predicting static liquefaction landslides.

Microstructure is a composite emergence of the state and nature of soils, which depends on basic state conditions, including packing, stress, and saturation, as well as inherent material properties such as grain size characteristics and mineral compositions. Thus, we use SEM to image microstructures in eight loess samples, which affords a direct path to analyze the effects of mechanical compaction, specimen disturbance, and soil nature on the shear behaviors of saturated loess, as shown in Fig. 13.

Loess is of the same nature as the Shichuan sampling site. Here, the naturally undisturbed loess (S_{1U}) has the largest pores and the loosest packing with a natural chain fabric, while the naturally remolded loess (S_{1R}) and compressively undisturbed loess (S_{2U}) exhibit denser packing with significantly fewer small pores at S_{2U} . Notably, it almost shows no pores in compressively remolded loess (S_{2R}), resulting in the densest packing due to the second reconstruction. Based on these observations, we explain the Effect of mechanical compaction and specimen disturbance on shear behaviors of saturated loess as follows. The naturally loose state produces a typical shear softening behavior on the saturated loess specimen (T_1 in Figs 4a-4c). The mechanical and manual disturbances-induced structure densification causes typical shear hardening behaviors in the saturated loess specimens ($T_2 \sim T_8$ in Figs. 4d-4f).

Meanwhile, the second reconstruction in S_{2R} results in much denser packing compared to S_{2U} , and consequently, the compressively remolded loess specimens ($T_6 \sim T_8$ in Figs. 4g-4i) have greater shear strength than the undisturbed specimens ($T_2 \sim T_5$ in Figs. 4d-4f). The creeping shear behaviors ($T_{11} \sim T_{13}$ in Fig. 5) at Yuzhong remolded loess (S_3) can contribute to the decrease in pore size and arrangement in fabric. The hardening shear behaviors (T_{14} in Fig. 5) on the same specimen are due to the higher confining pressure.

Loess is different from the other four sampling sites. There are two distinct types of structures. The remolded loess in Xiji (S_{4R}) has a relatively large grain size and pores, with the loosest packing, compared with Yan'an (S_{5R}), Xi'an (S_{6R}), and Tianshui (S_{7R}). Three remolded loess samples have denser packing due to fine grains filling pores and covering large grains. The Yan'an (S_{5R}) loess retains a certain amount of small pores. There is the finest-grained fraction in Tianshui (S_{7R}) loess without observed pores. There has also been no observed pore in Xi'an (S_{6R}), but its large grain fraction is between that of S_{5R} and S_{7R} . As a result, the remolded loess specimens at the Xiji sampling site (S_4) still exhibit shear softening behaviors even after undergoing manual disturbances ($T_{15} \sim T_{20}$ in Fig. 6), because they still retain relatively large pores and include more large grains, as shown in S_{4R} . The hardening shear behaviors ($T_{21} \sim T_{27}$ in Fig. 7) of the remolded loess specimen are related to the denser packing and finer grain at Yan'an (S_5), Xi'an (S_6), and Tianshui loess (S_7). Compared with Xi'an and Tianshui loess, Yan'an loess has greater shear strength ($T_{21} \sim T_{23}$ in Figs 7a-7c). It is reasonable to expect relatively larger grains and smaller pores, as shown in S_{5R} , S_{6R} , and S_{7R} .

On the other hand, loess from different sampling sites has distinct material properties. Fig. 14 shows a ternary diagram of the grain size characteristics of all loess in the dataset. It can be seen that there is a greater fine particle fraction in clayey and silty loess than in sandy loess. In addition,

previous studies have found that the clayey and silty loess comprises more clay minerals, such as interlayered illite/smectite, illite, and kaolinite (Zhang *et al.*, 2021c). Thus, inherent material properties only affect the shear behaviors of saturated loess (Fig. 14) and cannot obviously influence the extracted laboratory parameters (Figs. 8 and 9). This means that the shear strengths differ according to the shear behavior of saturated loess. Even if there exist some differences in the extracted laboratory parameters, the instability line and failure line still remain unique and similar under various stress and packaging states (Fig. 14). This is consistent with the conclusion on critical state soil mechanics (Schofield and Wroth, 1968; Been *et al.*, 1991). Of course, models based on laboratory parameters require careful analysis to ensure accurate prediction under in situ conditions.

4.2 Model adaptability for predicting static liquefaction landslides

To further investigate model adaptability, we conducted a sensitivity analysis of all parameter combinations in the developed sliding-block model. This is a crucial program for assessing model adaptability through sensitivity analysis. Fig. 15 illustrates the relative importance of each parameter in S_u and τ , that is, in Equations 1 and 5. The sensitivity analysis reveals the relative F_S value as a function of the relative value of the input parameters for a selected set of central values.

Generally, the less parameter variation and value variation, the lower the impact on a model. It can be seen that the sliding boundary parameters (a and z) and pore pressure parameter (r_u) have a more pronounced influence on F_S than the three extracted parameters (M , ϕ , and λ). Meanwhile, the three extracted parameters and the moisture condition parameter (γ_{sat}) have a minimal influence on F_S . When the central value of these parameters increases or decreases to the set maximum or minimum values, the F_S value and its variation do not exceed 0.2 and 20%, respectively. This first

means that a fully saturated assumption is reasonable in the developed sliding-block model. It is also crucial that this offers the most direct advantage for using the stress state framework to predict static liquefaction landslides. Because it is distinct from the strength laws in the Mohr–Coulomb model, which are also strongly dependent on the internal friction and cohesion of the soil, apart from thickness (z), slope (a), and pore pressure ratio (r_u) (Borga *et al.*, 2002).

4.3 Model advantage for predicting static liquefaction landslides

The sliding-block model, which integrates triaxial laboratory data, is based on the stress state framework for predicting static liquefaction landslides occurring in saturated loess. As it directly describes the internal distribution of force within a slope during external loading, for example, water infiltration-induced pore water pressure. In addition, it involves stress conditions and failure criteria, allowing for the convenient quantification of the internal stress acting along the sliding surface of a landslide in the Mohr circle. Based on results and analysis in the static liquefaction loess landslides (Figs 8 and 9), three key components need to be determined under stress state framework: (1) the failure criterion suitable for soil liquefaction; (2) the laboratory state parameters, here is M and λ , linking with the angle of shearing resistance and effective stress condition, respectively, and (3) the initial stress states, that is in-situ sliding boundary of a landslide, i.e., its thickness and slope.

Under the stress state framework, our sliding-block model is purely data-driven, relying solely on laboratory data. Meanwhile, its key state parameters are not only unaffected by the structure, state, and nature of the saturated loess, but also accurately predict both initiation and failure for all the static liquefaction loess landslides (Fig. 10). Moreover, the calculated pore water pressure can become a preferentially predictive tool for these loess landslides (Figs 8 and 9)

cooperated with field monitoring. Of course, the driven model's tests are not limited to undrained triaxial compression; other test methods can also be used as alternative programs, such as triaxial extension and direct simple shear (Sadrekarimi, 2020).

On the one hand, our model includes two key advantages. One is that the instability line and failure line are both unique, and the other is that the stress path spaces can extract parameters analogous to Mohr-Coulomb failure parameters (Fig. 3). Consequently, the parameters can easily be used in traditional mechanical equilibrium analysis on slopes. On the other hand, the foregoing results highlight the excellent performance of our model in predicting water-laden loess landslides at both laboratory and field scales (Fig. 5). Therefore, our model appears to be capable of accurately predicting static liquefaction landslides that occur in other soils.

4.4 Criterion for predicting future static liquefaction landslides

We develop a multi-index criterion for predicting future static liquefaction landslide (Fig. 16). In the loess example, the criterion also results from the two fitted equations in Figs. 9a and b, which directly link the force state and geometric boundary features (i.e., thickness and slope) of a landslide. Thus, it can easily be converted into a predictive method for the initiation and failure of static liquefaction landslides, using pore water pressure as a tool. Meanwhile, we use the results of static liquefaction loess landslides as an example to build a stability phase diagram (Fig. 16). In this example, when the factor of safety is less than 1, a given loess landslide will initiate or fail due to static liquefaction.

The stability phase diagram can provide different factors of safety against static liquefaction of a given landslide at various development stages, which directly depend on the pore

water pressure. Once M and λ are obtained from laboratory tests, together with instruments such as piezometers and groundwater level gauges installed on a landslide, they can produce a continuous monitoring input of pore pressure into our model. Theoretically, the developing laboratory data-driven sliding-block model can be used to predict field static liquefaction landslides that occurred in other soils. Our method can avoid a dense network of piezometer installations. However, according to the findings of our calculated results, it is best to have at least one piezometer installed near the sliding surface or zone. For groundwater fluctuations induced landslides, a water level gauge is essential to install on the landslide before obvious deformation occurs. Instruments like piezometers can gather real-time data. Finally, pore water pressure in landslides can provide vital signals in predicting static liquefaction events, when the in situ observed value is close to that predicted by a laboratory data-driven sliding-block model.

5. Conclusions

The shear behaviors of unsaturated loess in undrained triaxial compression tests showed three types of softening, creep, and hardening, which depend on the packing structure of the specimens. The compiled database of undrained triaxial compression tests of saturated loess combines current and published triaxial tests.

Our results show that the stress state framework is a preferential method in predicting the initiation and failure of static liquefaction landslides. The sliding-block model was developed by integrating the extracted state parameters M and λ into Mohr's circle failure criterion, which accurately predicted the factor of safety against initiation and failure of nine static liquefaction loess landslides. The predicted results showed that the stress state framework is a preferential

method in predicting the initiation and failure of static liquefaction landslides. The new sliding-block model is not only a model built on a faithful evolutionary basis, but also can make direct predictions for static liquefaction landslides. Thus, it can be used in other saturated soils.

The mode adaptability analysis revealed that the pore water pressure and boundary parameters in landslides are more sensitive to changes than the parameters extracted from the triaxial laboratory in the sliding-block model. The hydro-mechanical coupling criterion for predicting instability and failure of the static liquefaction landslides. Our findings demonstrate that the new sliding-block model bridges the gap between triaxial laboratory data and field site predictions, serving as an effective tool for predicting static liquefaction landslides within a stress state framework.

Acknowledgments

This study was supported by the National Natural Science Foundation of China (42090053), the National Key R&D Program of China (2022YFC3003401), and the Key project of Joint scientific Research Fund in Gansu Province (24JRRA814)

Data Availability Statement

The original data files in this study are freely available at Zhang (2025) via figshare (<https://doi.org/10.6084/m9.figshare.28388111.v1>).

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Ahenkorah, I., Rahman, M.M., Karim, M.R., Beecham, S., 2024. Cyclic liquefaction resistance of MICP- and EICP-treated sand in simple shear conditions: a benchmarking with the critical state of untreated sand. *Acta Geotechnica* 19, 5891–5913.
- Been, K., Jefferies, M.G., Hachey, J., 1991. The critical state of sands. *Géotechnique* 41, 365-381.
- Borga, M., Fontana, G.D., Gregoretti, C., Marchi, L., 2002. Assessment of shallow landsliding by using a physically based model of hillslope stability. *Hydrological Processes* 16, 2833-2851.
- Carraro, J.A.H., Prezzi, M., Salgado, R., 2009. Shear strength and stiffness of sands containing plastic or nonplastic fines. *Journal of Geotechnical and Geoenvironmental Engineering* 135, 1167-1178.
- Castro, G., 1969. Liquefaction of sands. Harvard University.
- Chen, G., Meng, X., Qiao, L., Zhang, Y., Wang, S., 2017. Response of a loess landslide to rainfall: observations from a field artificial rainfall experiment in Bailong River Basin, China. *Landslides* 15, 895-911.
- Chu, J., Wanatowski, D., 2012. Factors affecting pre-failure instability of sand under plane-strain conditions. *Géotechnique* 62, 121-135.
- Derbyshire, E., Dijkstra, T.A., Smalley, I.J., Li, Y., 1994. Failure mechanisms in loess and the effects of moisture content changes on remoulded strength. *Quaternary International* 24, 5-15.
- Feng, L., Zhang, S., Jin, Z., Zhang, M., Sun, P., Jia, J., Chu, G., Hu, W., 2021. The critical mechanics of the initiation of loess flow failure and implications for landslides. *Engineering Geology* 288, 106165.
- Ghafghazi, M., Shuttle, D.A., DeJong, J.T., 2014. Particle breakage and the critical state of sand. *Soils and Foundations* 54, 451-461.
- Gibbs, H.J., Holland, W.Y., 1960. Petrographic and engineering properties of loess. *Engineering Monographs* 28, 1-37.
- Holtz, R.D., Kovacs, W.D., Sheahan, T.C., 1981. An introduction to geotechnical engineering. Prentice Hall, Englewood Cliffs.
- Ishihara, K., 1993. Liquefaction and flow failure during earthquake. *Géotechnique* 43, 351-415.
- Ishihara, K., 2008. Flow slides of underwater sand deposits in Jamuna River bed. *Geotechnical Engineering for Disaster Mitigation and Rehabilitation: Proceedings of the 2nd International Conference GEDMAR08, Nanjing, China 30 May–2 June, 2008*. Springer, pp. 3-34.

Ishihara, K., Okusa, S., Oyagi, N., Ischuk, A., 1990. Liquefaction-induced flowslide in the collapsible loess deposit in Soviet Tajik. *Soils and Foundations* 30, 73-89.

Juang, C.H., Dijkstra, T., Wasowski, J., Meng, X., 2019. Loess geohazards research in China: Advances and challenges for mega engineering projects. *Engineering Geology* 251, 1-10.

Kong, J., Zhuang, J., Zhan, J., Bai, Z., Leng, Y., Ma, P., Peng, J., Wang, Z., Gu, T., Sun, J., Zhang, F., Xu, Y., Wang, Y., Mu, J., Fan, H., 2021. A landslide in Heifangtai, northwest of the Chinese Loess Plateau: triggered factors, movement characteristics, and failure mechanism. *Landslides* 18, 3407–3419.

Kramer, S.L., 1988. Triggering of liquefaction flow slides in coastal soil deposits. *Engineering Geology* 26, 17-31.

Kramer, S.L., Seed, H.B., 1988. Initiation of soil liquefaction under static loading conditions. *Journal of geotechnical engineering* 114, 412-430.

Lade, P.V., 1992. Static Instability and liquefaction of loose fine sandy slopes. *Journal of geotechnical engineering* 118, 51-71.

Lade, P.V., 1993. Initiation of static instability in the submarine Nerlerk berm. *Canadian Geotechnical Journal* 30, 895-904.

Leong, W.K., Chu, J., 2002. Effect of undrained creep on instability behaviour of loose sand. *Canadian Geotechnical Journal* 39, 1399-1405.

Li, Y., Shi, W., Aydin, A., Beroya-Eitner, M.A., Gao, G., 2020. Loess genesis and worldwide distribution. *Earth-Science Reviews* 201, 102947.

Liu, F., Xu, Q., Zhang, Y., Frost, J.D., Zhang, X., 2019. State-dependent flow instability of a silty loess. *Géotechnique Letters* 9, 22-27.

Liu, F., Xu, Q., Zhang, Y., Macedo, J.E., 2022. Influence of fines content on the undrained flow instability of loess. *Géotechnique Letters* 12, 1-7.

Lutenegger, A.J., 1981. Stability of loess in light of the inactive particle theory. *Nature* 291, 360-360.

Olson, S.M., Stark, T.D., 2003. Yield strength ratio and liquefaction analysis of slopes and embankments. *Journal of Geotechnical and Geoenvironmental Engineering* 129, 727-737.

Peng, J., Fan, Z., Wu, D., Zhuang, J., Dai, F., Chen, W., Zhao, C., 2015. Heavy rainfall triggered loess–mudstone landslide and subsequent debris flow in Tianshui, China. *Engineering Geology* 186, 79-90.

Peng, J., Wang, S., Wang, Q., Zhuang, J., Huang, W., Zhu, X., Leng, Y., Ma, P., 2019. Distribution and genetic types of loess landslides in China. *Journal of Asian Earth Sciences* 170, 329-350.

Peng, J., Zhuang, J., Wang, G., Dai, F., Zhang, F., Huang, W., Xu, Q., 2018. Liquefaction of loess landslides as a consequence of irrigation. *Quarterly Journal of Engineering Geology and Hydrogeology* 51, 330-337.

Poulos, S.J., Castro, G., France, J.W., 1985. Liquefaction evaluation procedure. *Journal of geotechnical engineering* 111, 772-792.

Qi, X., 2017. Sudden loess landslide monitoring and early warning research—a case study of Gansu landslide in Heifangtai Loess. *Chengdu University of Technology, Chengdu*.

Qi, X., Xu, Q., Liu, F., 2018. Analysis of retrogressive loess flowslides in Heifangtai, China. *Engineering Geology* 236, 119-128.

Reid, D., Dickinson, S., Mital, U., Fanni, R., Fourie, A., 2022. On some uncertainties related to static liquefaction triggering assessments. *Proceedings of the Institution of Civil Engineers - Geotechnical Engineering* 175, 181-199.

Reid, D., Fanni, R., Fourie, A., 2021. Discussion of "Forewarning of Static Liquefaction Landslides" by Abouzar Sadrekarimi. *Journal of Geotechnical and Geoenvironmental Engineering* 147, 07021020.

Sadrekarimi, A., 2020. Forewarning of static liquefaction landslides. *Journal of Geotechnical and Geoenvironmental Engineering* 146, 04020090.

Sadrekarimi, A., 2021. Closure to "Forewarning of Static Liquefaction Landslides" by Abouzar Sadrekarimi. *Journal of Geotechnical and Geoenvironmental Engineering* 147, 07021021.

Sadrekarimi, A., Olson, S.M., 2011. Yield strength ratios, critical strength ratios, and brittleness of sandy soils from laboratory tests. *Canadian Geotechnical Journal* 48, 493-510.

Schofield, A., Wroth, P., 1968. Critical state soil mechanics.

Scoppettuolo, M.R., Cascini, L., Babilio, E., 2020. Typical displacement behaviours of slope movements. *Landslides* 17, 1105-1116.

Seed, H.B., 1987. Design problems in soil liquefaction. *Journal of geotechnical engineering* 113, 827-845.

Shi, K., Zhu, F., Zhao, J., 2023. The critical state of crushable granular sand. *Acta Geotechnica*.

Sladen, J.A., D'Hollander, R.D., Krahn, J., 1985. The Liquefaction of Sands, a Collapse Surface Approach. *Canadian Geotechnical Journal* 26, 23-41.

Sun, P., Wang, H., Wang, G., Li, R., Zhang, Z., Huo, X., 2021. Field model experiments and numerical analysis of rainfall-induced shallow loess landslides. *Engineering Geology* 295, 106411.

Verdugo, R., Ishihara, K., 1996. The steady state of sandy soils. *soils and foundations* 36, 81-91.

Wang, G., Li, T., Xing, X., Zou, Y., 2015. Research on loess flow-slides induced by rainfall in July 2013 in Yan'an, NW China. *Environmental Earth Sciences* 73, 7933-7944.

Wang, G., Sassa, K., 2002. Post-failure mobility of saturated sands in undrained load-controlled ring shear tests. *Canadian Geotechnical Journal* 39, 821-837.

Wang, G., Sassa, K., Fukuoka, H., Tada, T., 2007. Experimental study on the shearing behavior of saturated silty soils based on ring-shear tests. *Journal of Geotechnical and Geoenvironmental Engineering* 133, 319-333.

Wang, Z.-L., Dafalias, Y.F., Li, X.-S., Makdisi, F.I., 2002. State pressure index for modeling sand behavior. *Journal of Geotechnical and Geoenvironmental Engineering* 128, 511-519.

Xian, Y., Wei, X., Zhou, H., Chen, N., Liu, Y., Liu, F., Sun, H., 2022. Snowmelt-triggered reactivation of a loess landslide in Yili, Xinjiang, China: mode and mechanism. *Landslides*.

Xu, J., Wei, W., Bao, H., Zhang, K., Lan, H., Yan, C., Sun, W., 2020a. Failure models of a loess stacked dam: a case study in the Ansai Area (China). *Bulletin of Engineering Geology and the Environment* 79, 1009-1021.

Xu, L., Coop, M.R., 2017. The mechanics of a saturated silty loess with a transitional mode. *Géotechnique* 0, 1-16.

Xu, L., Coop, M.R., Zhang, M., Wang, G., 2018. The mechanics of a saturated silty loess and implications for landslides. *Engineering Geology* 236, 29-42.

Xu, L., Dai, F.C., Tham, L.G., Zhou, Y.F., Wu, C.X., 2012. Investigating landslide-related cracks along the edge of two loess platforms in northwest China. *Earth Surface Processes and Landforms* 37, 1023-1033.

Xu, L., Gao, C., Zuo, L., Liu, K., Li, L., 2022. The critical states of saturated loess soils. *Engineering Geology* 307, 106776.

Xu, Q., Peng, D., Zhang, S., Zhu, X., He, C., Qi, X., Zhao, K., Xiu, D., Ju, N., 2020b. Successful implementations of a real-time and intelligent early warning system for loess landslides on the Heifangtai terrace, China. *Engineering Geology* 278, 105817.

Yang, J., 2002. Non-uniqueness of flow liquefaction line for loose sand. *Géotechnique* 52, 757-760.

Yang, J., Liang, L., Chen, Y., 2022. Instability and liquefaction flow slide of granular soils: the role of initial shear stress. *Acta Geotechnica* 17, 1-15.

Yoshimine, M., Ishihara, K., 1998. Flow potential of sand during liquefaction. *Journal of the Japanese Geotechnical Society : soils and foundation* 38, 189-198.

Zhang, D., Wang, G., 2007. Study of the 1920 Haiyuan earthquake-induced landslides in loess (China). *Engineering Geology* 94, 76-88.

Zhang, D., Wang, G., Luo, C., Chen, J., Zhou, Y., 2009. A rapid loess flowslide triggered by irrigation in China. *Landslides* 6, 55-60.

Zhang, F., Huang, X., 2018. Trend and spatiotemporal distribution of fatal landslides triggered by non-seismic effects in China. *Landslides* 15, 1663–1674.

Zhang, F., Peng, J., Huang, X., Lan, H., 2021a. Hazard assessment and mitigation of non-seismically fatal landslides in China. *Natural Hazards* 106, 785-804.

Zhang, F., Peng, J., Wu, X., Pan, F., Jiang, Y., Kang, C., Wu, W., Ma, W., 2021b. A catastrophic flowslide that overrides a liquefied substrate: the 1983 Saleshan landslide in China. *Earth Surface Processes and Landforms* 46, 2060-2078.

Zhang, F., Wang, G., 2018. Effect of irrigation-induced densification on the post-failure behavior of loess flowslides occurring on the Heifangtai area, Gansu, China. *Engineering Geology* 236, 111-118.

Zhang, F., Wang, G., Kamai, T., Chen, W., 2014. Effect of pore-water chemistry on undrained shear behaviour of saturated loess. *Quarterly Journal of Engineering Geology and Hydrogeology* 47, 201-210.

Zhang, F., Wang, G., Kamai, T., Chen, W., Zhang, D., Yang, J., 2013. Undrained shear behavior of loess saturated with different concentrations of sodium chloride solution. *Engineering Geology* 155, 69-79.

Zhang, F., Wang, G., Peng, J., 2022. Initiation and mobility of recurring loess flowslides on the Heifangtai irrigated terrace in China: Insights from hydrogeological conditions and liquefaction criteria. *Engineering Geology* 302, 106619.

Zhang, F., Yan, B., Feng, X., Lan, H., Kang, C., Lin, X., Zhu, X., Ma, W., 2019. A rapid loess mudflow triggered by the check dam failure in a bulldoze mountain area, Lanzhou, China. *Landslides* 16, 1981-1992.

Zhang, F., Zhao, C., Lourenço, S.D.N., Dong, S., Jiang, Y., 2021c. Factors affecting the soil–water

retention curve of Chinese loess. *Bulletin of Engineering Geology and the Environment* 80, 717-729.

Zhou, J., Zhu, C., Zheng, J., Wang, X., Liu, Z., 2002. Landslide disaster in the loess area of China. *Journal of Forestry Research* 13, 157-161.

Zhou, W., Qiu, H., Wang, L., Pei, Y., Tang, B., Ma, S., Yang, D., Cao, M., 2022. Combining rainfall-induced shallow landslides and subsequent debris flows for hazard chain prediction. *CATENA* 213, 106199.

Zhuang, J., Peng, J., 2014. A coupled slope cutting—a prolonged rainfall-induced loess landslide: a 17 October 2011 case study. *Bulletin of Engineering Geology and the Environment* 73, 997-1011.

Zhuang, J., Peng, J., Du, C., Zhu, Y., Kong, J., 2024. Shallow-landslide stability evaluation in loess areas according to the Revised Infinite Slope Model: a case study of the 7.25 Tianshui sliding-flow landslide events of 2013 in the southwest of the Loess Plateau, China. *Natural Hazards and Earth System Sciences* 24, 2615-2631.

Zhuang, J., Peng, J., Wang, G., Iqbal, J., Wang, Y., Li, W., Xu, Q., Zhu, X., 2017. Prediction of rainfall-induced shallow landslides in the Loess Plateau, Yan'an, China, using the TRIGRS model. *Earth Surface Processes and Landforms* 42, 915-927.

Figure captions

Fig. 1 The grain size curves and Atterberg limits of testing loess at different sampling sites

Fig. 2 Schematic maps of four shear behaviors of saturated loess. (a) Stress-strain curves; (b) Pore pressure-strain curves; (c) Effective stress paths.

Fig. 3 Schematic map of sliding-block model with a graphical procedure in Mohr circle.

Fig. 4 The Effect of mechanical compaction on shear behaviors of saturated loess specimens at Shichuan (S_1 and S_2) sampling site. (a) Stress-strain curves; (b) Pore pressure-strain curves; (c) Effective stress paths.

Fig. 5 The Effect of disturbance on shear behaviors of saturated loess specimens at Yuzhong (S_3) sampling site. (a) Stress-strain curves; (b) Pore pressure-strain curves; (c) Effective stress paths.

Fig. 6 The Effect of disturbance on shear behaviors of saturated silty loess specimens at Xiji (S_4) and Yan'an (S_5) sampling sites. (a) Stress-strain curves; (b) Pore pressure-strain curves; (c) Effective stress paths.

Fig. 7 The Effect of disturbance on shear behaviors of saturated clayey loess specimens at Xi'an (S_6) and Tianshui (S_7) sampling sites. (a) Stress-strain curves; (b) Pore pressure-strain curves; (c) Effective stress paths.

Fig. 8 State lines in mean effective principal stress (p') and deviator stress (q) space at instability and failure points of all the saturated loess samples. (a) Instability points; (b) Failure points.

Fig. 9 The relationship of pore water pressure ratio (r_u) and normalized stress ratio (p'/σ'_{3c}) at instability and failure points of all the saturated loess samples. (a) Instability points; (b) Failure points.

Fig. 10 The factor of safety against static liquefaction loess landslides from six flume tests and three field monitoring events. (a) The factor of safety against initiation of static liquefaction (b)

The factor of safety against failure of static liquefaction.

Fig. 11 Model predicting results at initiation and failure for nine static liquefaction loess landslides.

(a) and (b) Comparison of predicted and observed r_u ; (c) and (d) Comparison of predicted and observed T_N .

Fig. 12 Time-lapsed interval (T_L) from initiation to failure.

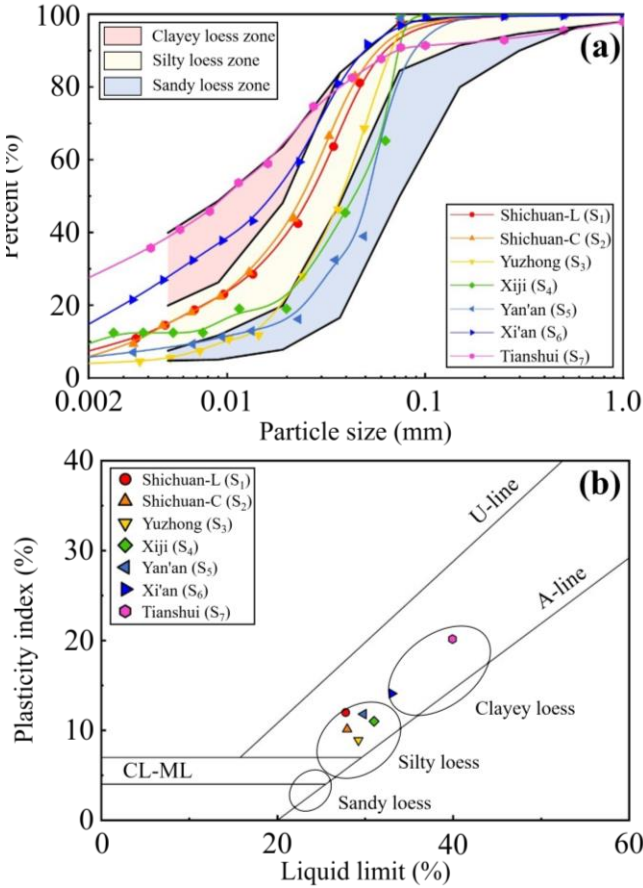
Fig. 13 SEM images of the six loess sampling sites.

Fig. 14 Ternary diagram of grain size characteristic all loess on the dataset. Note. The descriptions of the soils use the classification suggested by the USDA' taxonomy (Soil Survey Division Staff, 1993); viz. sand: 0.05–2 μm , silt: 0.05–0.002 μm , clay: <0.002 μm .

Fig. 15 Sensitivity diagram of all parameters in the developed sliding-block model. (a) Extrated parameters (M , ϕ , and λ); (b) Sliding boundary and state parameters (γ_{sat} , a , z , r_u).

Fig. 16 A stability phase diagram for the factor of safety against static liquefaction landslides. (a) Exemplified initiation criterion in loess landslides; (b) Exemplified failure criterion in loess landslides.

794 **Fig. 1**



795

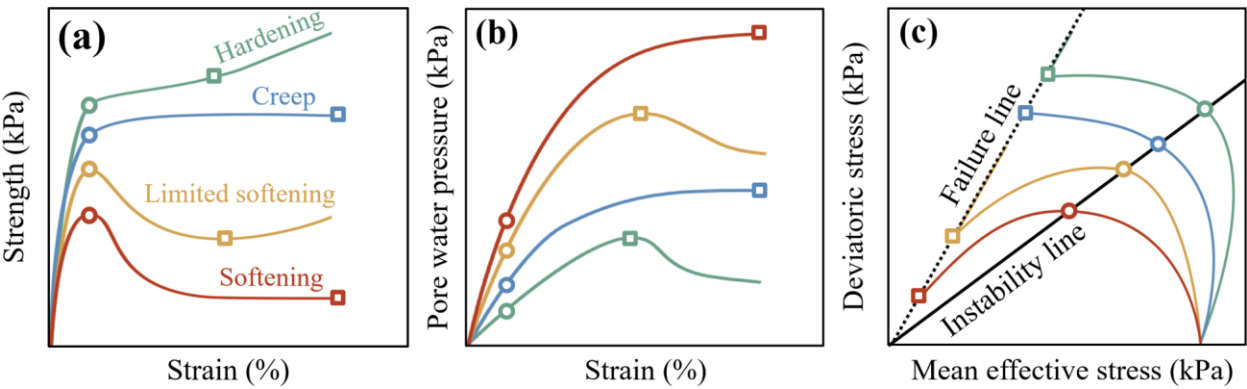
796

797

798

799

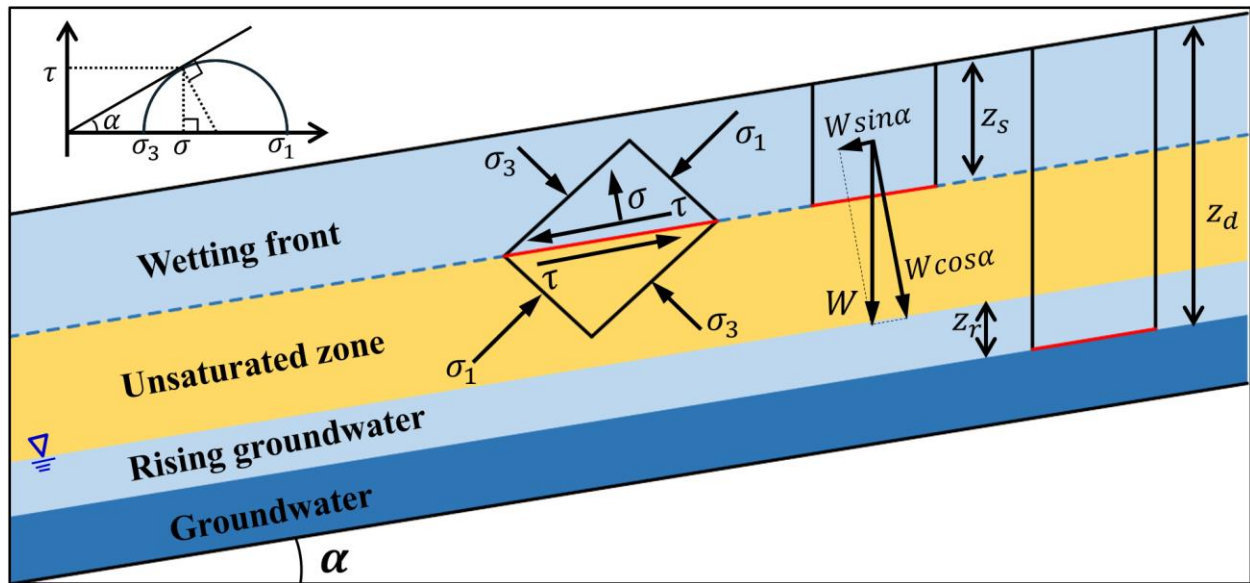
800 **Fig. 2**



801

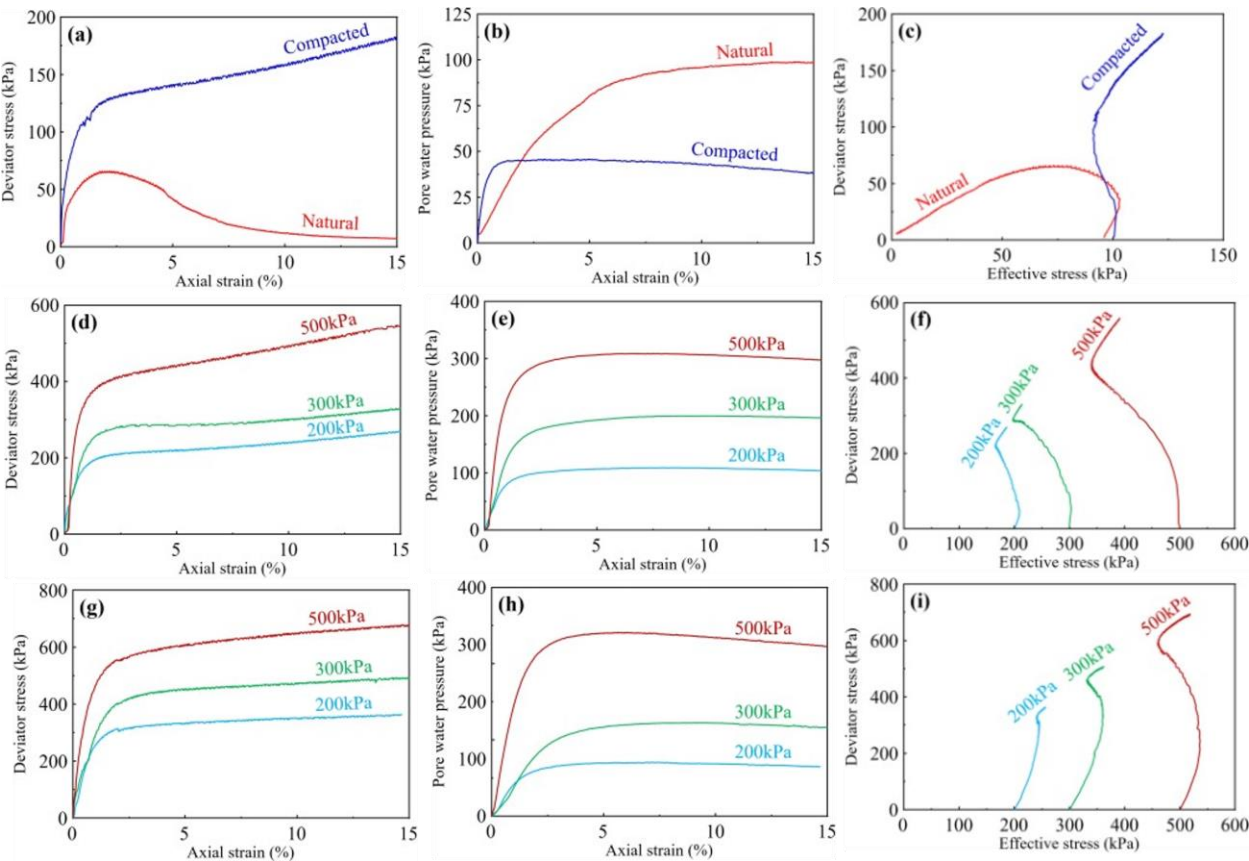
802

Fig. 3



807

808 **Fig. 4**

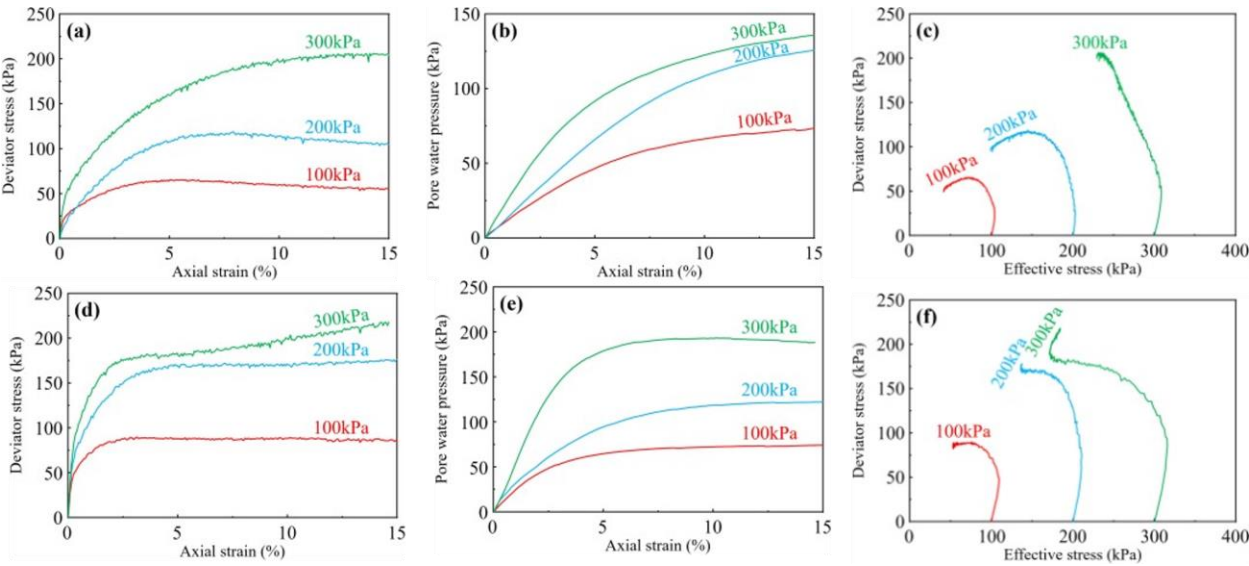


809

810

811

812 **Fig. 5**



813

814

Fig. 6

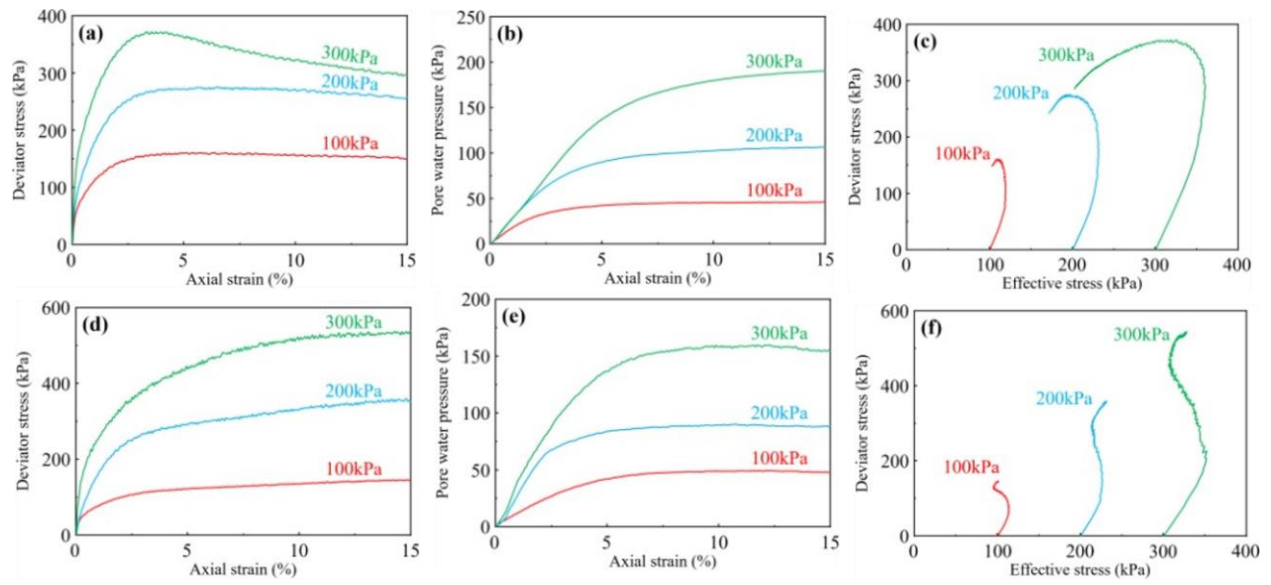


Fig. 7

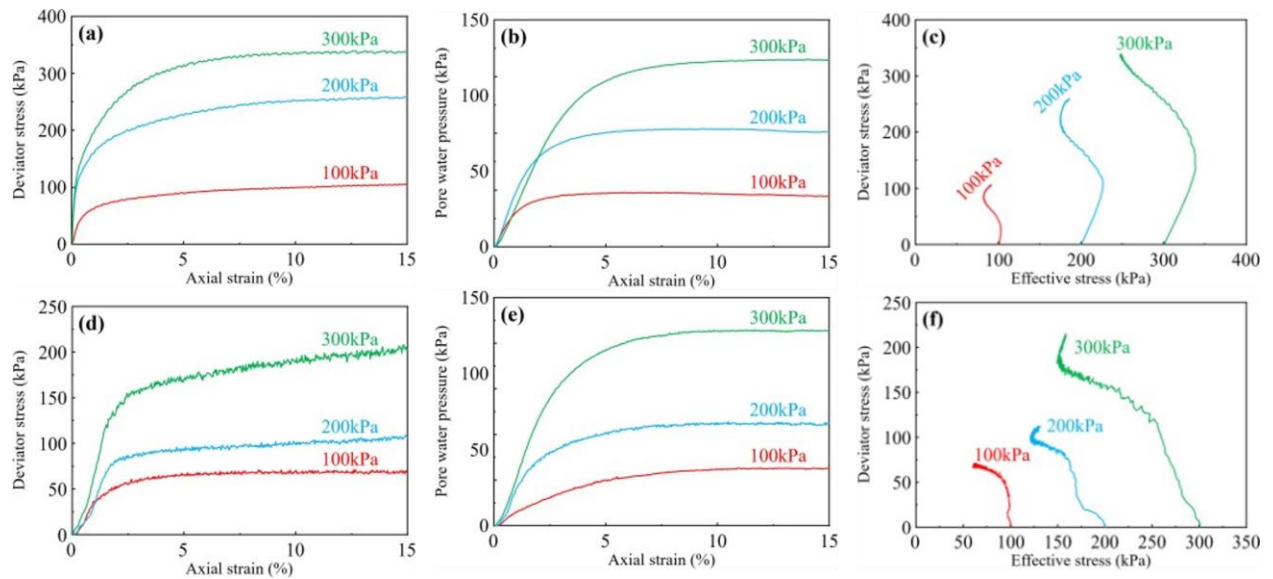


Fig. 8

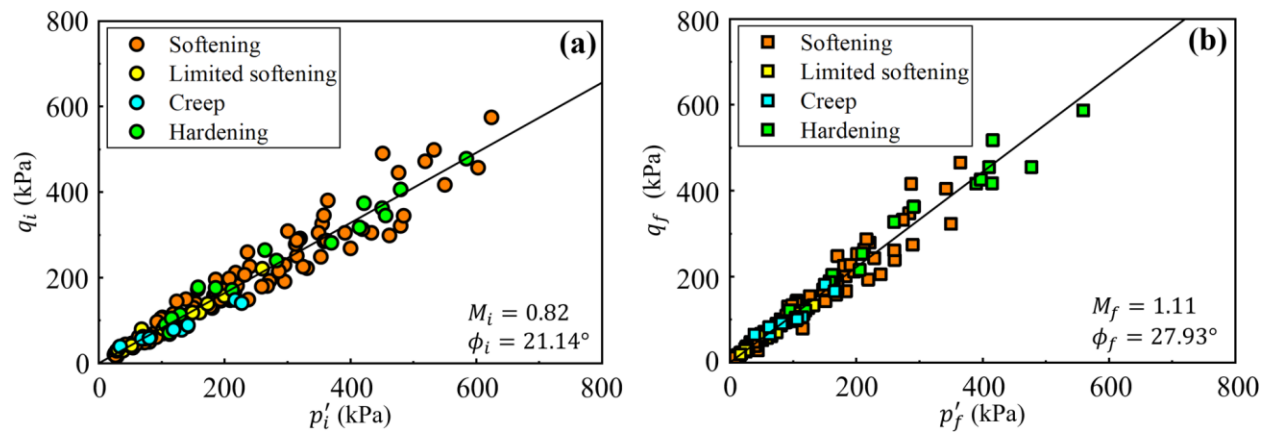


Fig. 9

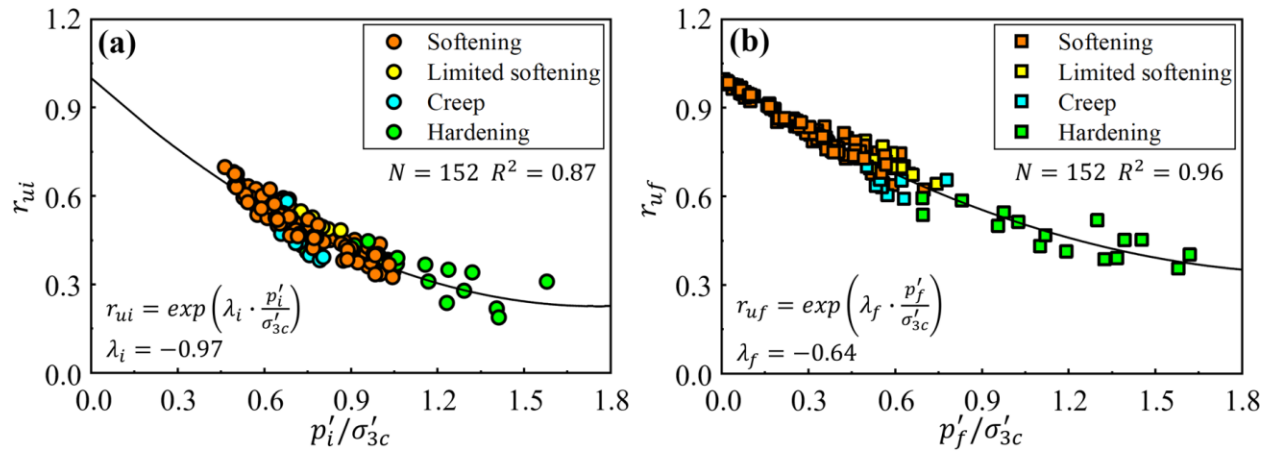


Fig. 10

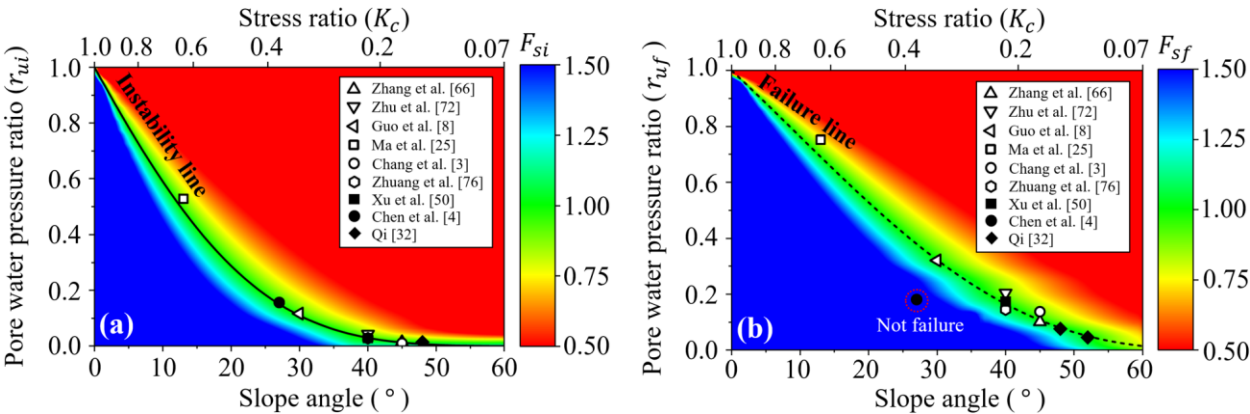
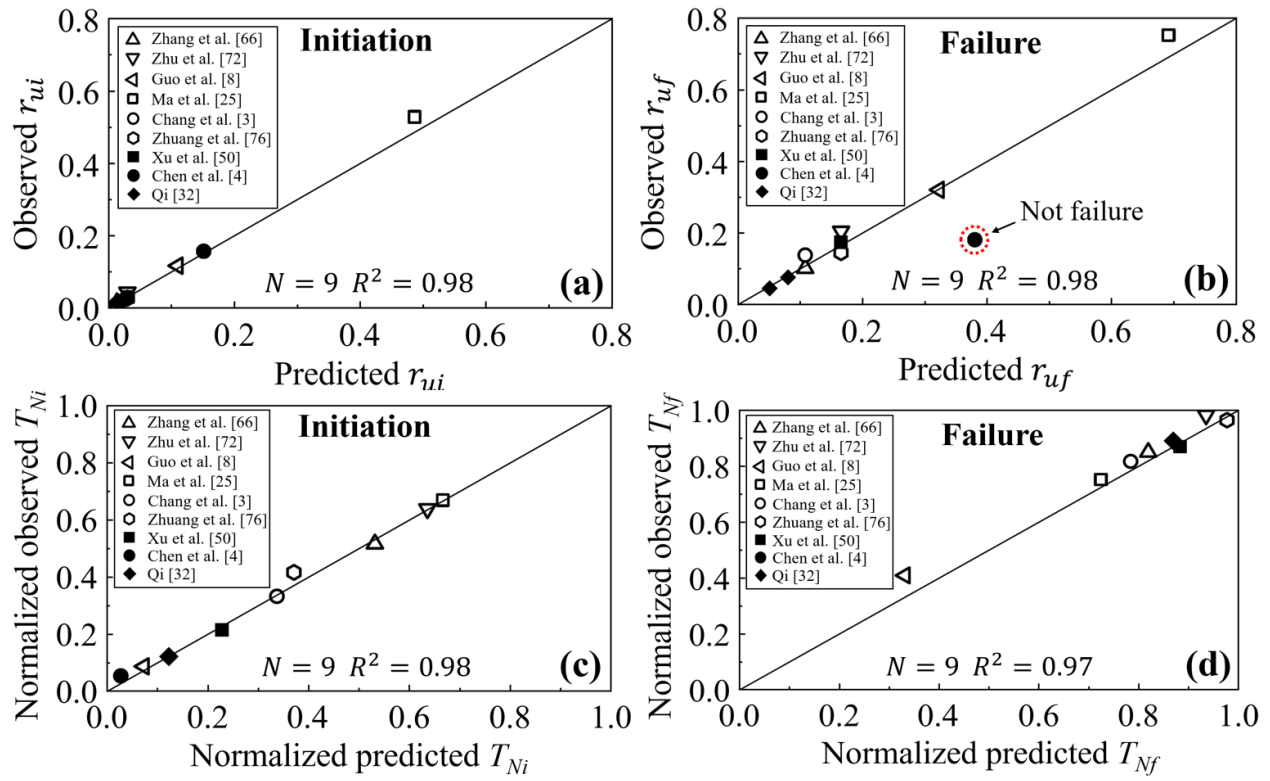
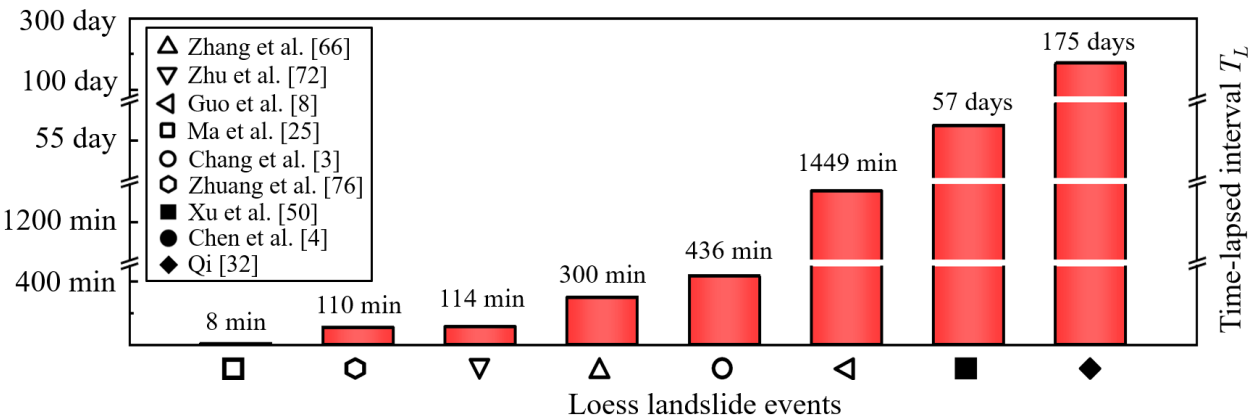


Fig.11



842

843 **Fig. 12**



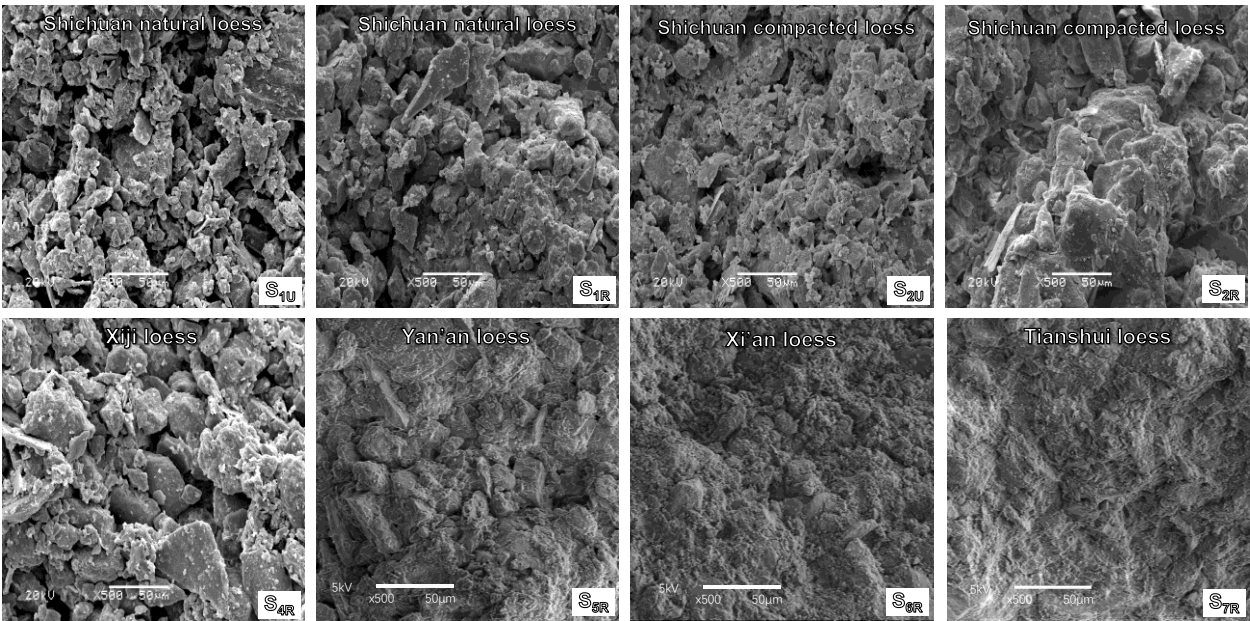
844

845

846

847

848 **Fig. 13**

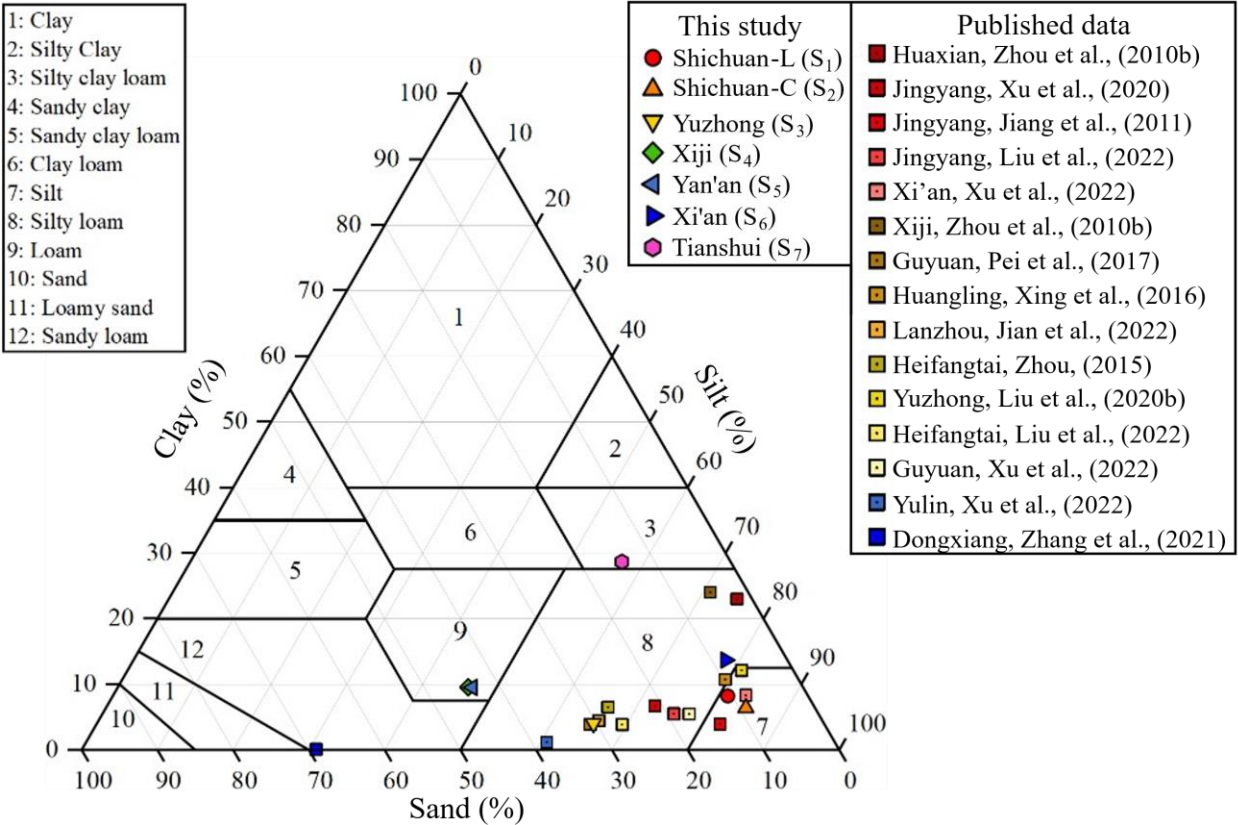


849

850

851

852 **Fig. 14**



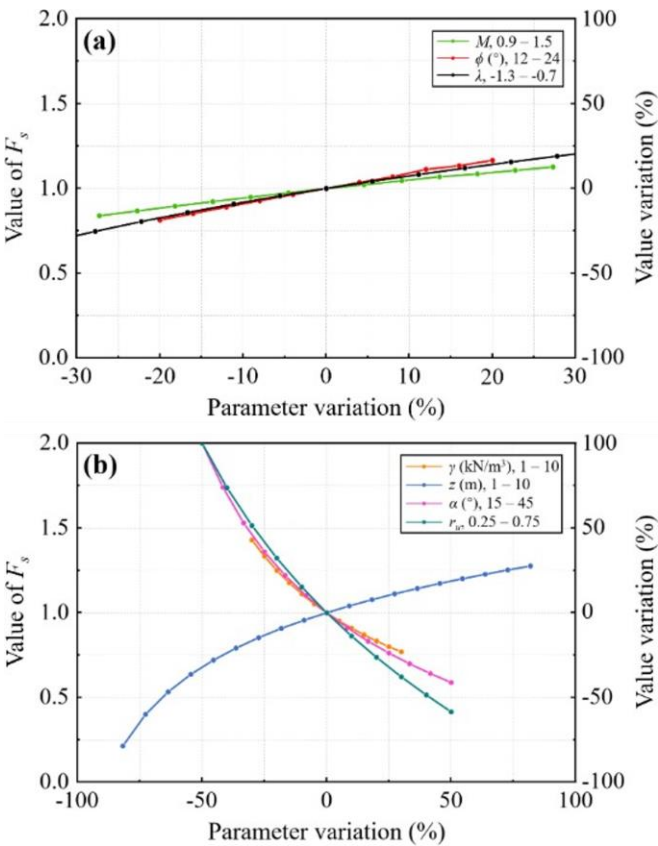
853

854

855

856

857 **Fig. 15**

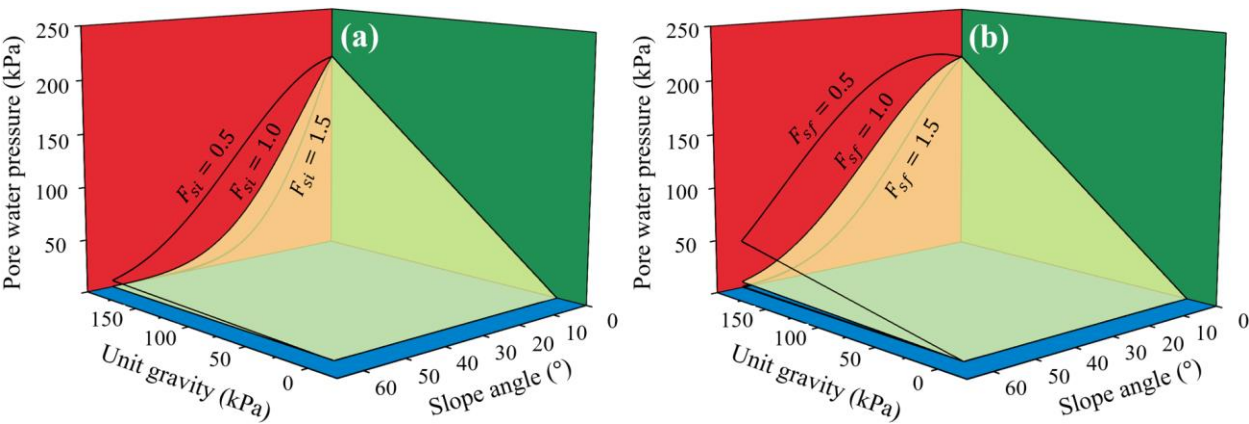


858

859

860

861 **Fig. 16**



862

863

864 **Table captions**

865 **Table. 1** The basic index properties of all the samples

866 **Table. 2** Summary of triaxial compression tests of saturated loess

867

Table. 1

SN	Site	Loess type	Basic properties				Grain parameters			
			G_s	ρ_d	w_L	w_P	I_P	C_u	C_c	D_{50}
S ₁	Shichuan-L	Silty	2.71	1.39	27.8	15.8	12.0	11.03	2.08	0.026
S ₂	Shichuan-C	Silty	2.71	1.79	27.9	17.8	10.1	8.56	1.76	0.024
S ₃	Yuzhong	Silty	2.70	1.42	29.2	20.3	8.9	4.39	1.37	0.038
S ₄	Xiji	Silty	2.71	1.50	31.0	20.0	11.0	25.81	5.89	0.044
S ₅	Yan'an	Silty	2.70	1.50	29.8	18.0	11.8	25.57	5.95	0.044
S ₆	Xi'an	Clayey	2.71	1.50	33.0	18.9	14.1	16.43	1.04	0.017
S ₇	Tianshui	Clayey	2.71	1.50	39.9	19.7	20.2	22.71	0.56	0.010

SN: sampling number, G_s : specific gravity, ρ_d : initial dry density (g/cm³), w_L : liquid limit (%), w_P : plastic limit (%), I_P : plastic index (%), C_u : uniformity coefficient, C_c : coefficient of curvature, and D_{50} : average grain diameter (mm).

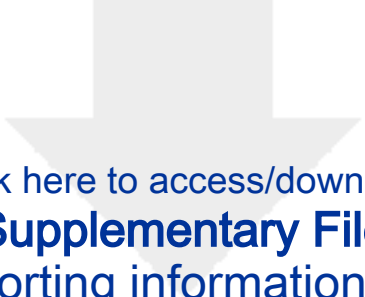
Note: SN is sampling number, Shichuan-L is a natural loess, and Shichuan-C is pseudo-natural loess with compaction in a land development project; Loess type is based on grain size curves and Atterberg limits

879
880

Table. 2

Number		Testing condition			State value at instability			State value at failure			Behavior
S	T	S_c	ρ_d	σ_{3c}	r_{ui}	M_i	ϕ_i	r_{uf}	M_f	ϕ_f	
Effect of mechanical compaction											
S ₁	T ₁	U	1.39	100	0.47	0.88	22.6	0.98	1.38	34.1	S
S ₂	T ₂ [*]	U	1.79	100	0.33	0.82	21.1	0.45	1.27	31.6	H
	T ₃ [*]	U	1.79	200	0.33	0.86	22.1	0.54	1.32	32.8	H
	T ₄ [*]	U	1.79	300	0.4	0.88	22.6	0.62	1.37	33.9	H
	T ₅ [*]	U	1.79	500	0.44	0.87	22.3	0.62	1.34	33.2	H
	T ₆	R	1.79	200	0.18	0.95	24.2	0.35	1.40	34.6	H
	T ₇	R	1.79	300	0.24	0.89	22.8	0.38	1.35	33.4	H
	T ₈	R	1.79	500	0.4	0.88	22.6	0.53	1.34	33.2	H
Effect of specimen disturbance											
S ₃	T ₉	U	1.42	100	0.3	0.81	22.6	0.61	1.29	30.7	S
	T ₁₀	U	1.42	200	0.44	0.61	20.4	0.57	0.91	26.1	S
	T ₁₁	U	1.42	300	0.32	0.64	17.1	0.64	1.09	28.4	C
	T ₁₂	R	1.42	100	0.48	0.88	20.9	0.75	1.23	32.1	C
	T ₁₃	R	1.42	200	0.46	0.79	16.1	0.68	1.03	23.3	H
	T ₁₄	R	1.42	300	0.33	0.65	16.8	0.58	1.13	27.5	H
Effect of soil nature											
S ₄	T ₁₅	R	1.50	100	0.24	1.04	26.3	0.47	1.44	35.5	S
	T ₁₆	R	1.50	200	0.27	1.05	26.5	0.54	1.41	34.8	S
	T ₁₇	R	1.50	300	0.25	1.03	26.1	0.64	1.42	35.0	S
S ₅	T ₁₈	R	1.50	100	0.23	0.91	23.3	0.41	1.22	30.5	H
	T ₁₉	R	1.50	200	0.24	0.95	24.2	0.39	1.31	32.5	H
	T ₂₀	R	1.50	300	0.22	0.94	24.0	0.42	1.33	33.0	H
S ₆	T ₂₁	R	1.50	100	0.36	0.78	20.2	0.48	1.13	28.4	H
	T ₂₂	R	1.50	200	0.21	0.72	18.8	0.5	1.29	32.1	H
	T ₂₃	R	1.50	300	0.21	0.70	18.3	0.54	1.32	32.8	C
S ₇	T ₂₄	R	1.50	50	0.32	0.84	21.6	0.48	1.16	29.1	C
	T ₂₅	R	1.50	100	0.3	0.63	16.6	0.6	1.07	27.0	H
	T ₂₆	R	1.50	200	0.48	0.61	16.1	0.63	1.02	25.9	H
	T ₂₇	R	1.50	300	0.40	0.63	16.6	0.68	1.12	28.2	H

881 Note: S: sampling number, T: testing number, *: published data (Zhang *et al.*, 2019a); S_c : testing
882 sample condition, U: undisturbed samples, R: remolded samples; ρ_d : dry density (g/cm³), σ_{3c} :
883 confining pressure (kPa); r_u : pore water pressure ratio, M : stress ratio ($= q/p'$), and ϕ : the angle of
884 shearing resistance (°); S: softening, L: limited softening, C: creep, H: hardening.



Click here to access/download
Supplementary File
Supporting information.docx

